

UNDERSTANDING TIME IN NATURAL LANGUAGE

PH.D. THESIS PROPOSAL (ABSTRACT)

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Abstract

Time is an important dimension when we describe the world because the world is evolving over time and many facts are time-sensitive. Temporal understanding is thus an important aspect of natural language understanding and many applications may rely on it, e.g., information retrieval, summarization, causality, and question answering.

My thesis research has so far focused on temporal relation extraction, a key component of temporal understanding. That is, given a pair of events or actions, determine which of them occurred first (or other temporal relations between them, e.g., simultaneous or overlapping). The task is challenging because the actual timestamps of those events or actions are rarely expressed explicitly and their temporal order has to be inferred, from lexical cues, between the lines, and often based on strong background knowledge. The task becomes even more challenging when multiple events or documents are considered and a globally coherent temporal structure is expected.

My contribution has been along two directions. One is the *machine learning direction*, where I propose to better exploit the structure induced by the transitivity property of temporal relations in the structured learning framework [1-2]; the other one is from the *natural language perspective*, where I investigate how temporal relations are expressed (from the authors' standpoint) and perceived (from the readers' standpoint) [3-5]. *Integrating these two improvements, my current system significantly improves the state-of-the-art temporal relation extraction performance by approximately 20% in F_1 .* More technical details will be provided in my thesis proposal.

In my future thesis work, I propose to continue my study of temporal information extraction and pursue two key directions. **First**, I will continue improving temporal relation extraction in the following ways: (1) Extend the dataset and knowledge base I generated in [3,5] to cover nominal events as well. (2) Incorporate an event coreference component so that long-distance events can be related. (3) Extend my current technique to multi-axis events (as defined in [5]) so that a more complete temporal structure can be captured. **Second**, I plan to move forward to multi-document temporal understanding (e.g., timeline generation). This will be practically more useful, but also technically more interesting/challenging due to its requirement for a synergy of entity coreference, event coreference, and temporal relation extraction methods, all in the cross-document setting that is known to be more difficult.

1. Q. Ning, Z. Feng, and D. Roth, "A Structured Learning Approach to Temporal Relation Extraction," EMNLP, 2017. [link]
2. Q. Ning, Z. Yu, C. Fan, and D. Roth, "Exploiting Partially Annotated Data in Temporal Relation Extraction," *SEM, 2018. [link]
3. Q. Ning, H. Wu, H. Peng, and D. Roth, "Improving Temporal Relation Extraction with a Globally Acquired Statistical Resource," NAACL, 2018. [link]
4. Q. Ning, Z. Feng, and D. Roth, "Joint Reasoning for Temporal and Causal Relations," ACL, 2018. [link]
5. Q. Ning, H. Wu, and D. Roth, "A Multi-Axis Annotation Scheme for Event Temporal Relations," ACL, 2018. [link]